

**VIRGINIA
MASS CASUALTY INCIDENT
MANAGEMENT**

**Module I
AWARENESS LEVEL
Participant Manual**



**6th Edition
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Office of Emergency Medical Services
Virginia Department of Health
Richmond, Virginia**

MASS CASUALTY INCIDENT MANAGEMENT

Module I

AWARENESS LEVEL

6th Edition

I. COURSE INFORMATION

A. Course Objectives

1. On completion of the course students will be able to:
 - i. Define Mass Casualty Incident
 - ii. List the three goals of mass casualty incident management
 - iii. Describe initial response actions and medical management of mass casualty incidents
 - iv. Understand the difference between START and SALT triage methods
 - v. Triage simulated patients correctly using SALT and START algorithms
 - vi. Understand the critical need for casualty accountability
 - vii. Review Virginia Triage tags

B. Overview

1. Virginia recognizes that there are multiple methods for triaging patients at a mass casualty incident. Historically, START has been the primary triage method taught and recognized in Virginia. The development of a national triage standard facilitated the need to introduce the SALT triage concept.
2. The Mass Casualty Incident Management program is designed to provide an overview of the structure and layout of the National Incident Management System.
3. While there are multiple components of the NIMS system, this course focuses on the medical component of NIMS.
4. For additional training in NIMS, take the following courses:
 - i. IS 100 – Introduction to Incident Command System
 - ii. IS 200 – ICS for Single Resource and Initial Action Incidents
 - iii. IS 700 – An Introduction to National Incident Management System
 - iv. IS 800 – An Introduction to National Response Framework

II. MASS CASUALTIES

A. Understanding mass casualties

1. Virginia's geographic location, areas of dense population, major transportation routes and unique hazards provide the potential for incidents to occur which could result in significant numbers of casualties, overwhelming available local resources.
2. Many people confuse a Mass Casualty Incident with a Disaster. As emergency response personnel it is important to understand what a disaster is and why most Mass Casualty Incidents are not true disasters.
3. A disaster declaration is made by the President of the United States and entitles the locality to federal aid. Disaster declarations come as the result of significant incidents that cripple or destroy large portions of the infrastructure of a state or locality. A disaster may or may not generate a mass casualty incident.
4. Examples include:
 - a. Natural (flood, hurricanes)
 - b. Technical hazards (HAZMAT, building collapse)
 - c. Transportation
 - d. Civil and political disorder
 - e. Criminal or terrorist incidents
 - f. Large Scale event
 - g. Pandemic

B. Mass Casualty Incidents

1. A mass casualty incident (or MCI) is any incident that injures enough people to overwhelm the available resources of a particular system or area.
 - a. It is important to remember that an MCI resulting from any type of manmade attack is considered a crime. Personnel should take care to preserve the crime-scene to the best of their abilities without compromising the ability to treat patients.
2. Goals of MCIM
 - a. Do the greatest good for the greatest number
 - i. Remember that heroic resuscitative efforts are not appropriate because they:

- (a) Take too much time
 - (b) Take equipment from salvageable patients
 - (c) Personnel intensive
 - ii. Concentrate on salvageable patients
- b. Manage scarce resources
 - i. Providers must remember that MCIs place a great demand on resources, including equipment, rescuers, and facilities.
- c. Do not relocate the disaster
 - i. Prioritize patients following the appropriate triage methodology
 - ii. Transport patients to all available hospitals instead of overloading the closest facility
 - iii. It is possible that some victims will self-transport, significantly impacting the facility's ability to accept additional patients.

III. EMS INITIAL RESPONSE ROLES AND RESPONSIBILITIES

A. Initial response roles

1. EMS is a specific component of the overall incident management system. The first arriving unit should start the following actions.
2. The steps associated with taking control of a scene and initiating incident command are presented in this program as individual steps, however some of them may occur in conjunction with each other. For instance, it is nearly impossible to conduct the first two steps (Safety and Survey) as separate steps.

B. Establish Command

1. The first emergency response unit to arrive at a mass casualty incident is by default Incident Commander until relieved. As a result, the individuals on the first arriving unit must take immediate actions to begin to manage the entire incident. These actions may be the most important steps taken in the entire incident.
2. The initial unit must resist the "temptation" to begin one-on-one patient care.

C. The Five Ss

1. S-1: Safety:

- a. The first arriving unit places priority on scene safety
- b. Declare a MCI and advise incoming units to stage at a specific location until the scene is secure or they are advised to come to the scene.
- c. Identify specific hazards and mitigate them, or call for the necessary resources to render the scene safe.

2. S-2: Size-up:

- a. Identify the type of incident
- b. Determine an approximate number of patients
- c. Determine the severity and type of injuries or illnesses
 - i. Understanding the types of injuries allows responders time to ensure appropriate facilities are available and prepared
- d. Determine the area involved and the best scene access and egress

3. S-3: Send Information:

- a. The first arriving unit should immediately report the situation by providing an initial situation report to the dispatch center.
- b. Additional resources should be requested as needed. Be specific in the request, making sure that the necessary numbers and capabilities of resources are understood. This can include requesting aeromedical resources as well.
- c. Ensure that local hospitals are notified of the incident. Local protocols should be followed in notification of a Regional Hospital Coordination Center (RHCC) or command hospital.
- d. Early notification of the Medical Examiner is also essential when there are deceased patients.

4. S-4: Set-up:

- a. Providers should set-up the scene for the best management of mass casualties by on-scene and responding resources, including:
 - i. Incident Command Structure
 - ii. Staging
 - iii. Secure Access and Egress
 - iv. Secure adequate space for work areas including triage, treatment, and transportation

5. S-5: Start Triage:

- a. Triage ensures patients are rapidly assessed and sorted so that those with the most significant injuries and probability of survival receive treatment and transportation effectively
- b. There are multiple triage systems in use around the world. It is important to know what your agency/jurisdiction uses and what your mutual aid agencies use so that you can integrate appropriately.
- c. START and SALT are the two triage systems recognized in Virginia

IV. TRIAGE

A. Purpose:

- 1. Assigns treatment priorities.
- 2. Separates MCI victims into easily identifiable groups.
- 3. Determine required resources for treatment, transportation, and definitive care.
- 4. Prioritization of patient distribution and transportation.

B. Benefits:

- 1. Identifies patients who require rapid medical care to save life and limb.
- 2. Provides rational distribution of casualties.
- 3. Separating out the minor injuries reduces the urgent burden on each hospital
 - a. On average 10-15% of MCI patients are serious enough to require extended hospitalization.

C. The ideal triage system:

1. Should be simple.
2. Should be easy to perform
3. Should be easy to teach and learn
4. Provides for rapid and simple life-saving interventions
5. Does not require a specific diagnosis
6. Is part of an ongoing process

V. START TRIAGE

A. General Information

1. Categorizes patients into universal color categories based upon a rapid assessment
2. Triage ribbons made of surveyor's tape are used to mark patients

B. Categories

1. RED

- a. Patients categorized as RED are considered the highest priority
- b. Typical problems are:
 - i. R – Respirations/airway
 - ii. P – perfusion/pulse
 - iii. M – mental status
 - iv. Severe burns which compromise airways.

2. YELLOW

- a. Patients categorized as YELLOW are the second priority
- b. Typical problems are:
 - i. Burn patients without airway problems
 - ii. Major or multiple bone or joint injuries
 - iii. Back and spine injuries.

3. GREEN

- a. Patients categorized as GREEN are the third priority
- b. Typical problems are:
 - i. “Walking wounded”
 - ii. The ability to “walk” does not necessarily mean that this is a “minor” patient.
 - iii. Minor cuts and bruises are acceptable criteria for this type of patient.
 - iv. Minor painful swollen deformities
 - v. Minor soft tissue injuries.

4. BLACK

- a. Black tagged patients are considered dead/non-salvageable and are for that reason the lowest priority
- b. Black tagged patients are non-breathing individuals on whom resuscitation would normally be attempted but who are not salvageable given the resources available early in a MCI response.

C. Minimal Treatment

- 1. In an effort to ensure that responders do not get too focused on a single patient, only three treatments occur during the initial triage phase.
- 2. The minimal treatments are:
 - a. Opening the airway
 - b. Stopping major bleeding
 - c. Placing the patient in a safe position
 - i. Shock position
 - ii. Recovery position

V. **START TRIAGE**

A. Basic Guidelines

1. Move in an orderly pattern
 - a. Assess each casualty you come to and mark the priority using triage ribbons.
2. Maintain a patient count
 - a. Mark each patient tagged on 2-3 inch tape on thigh.
 - b. Responders can also save a small piece of triage ribbon and count each piece once tagging is completed.
3. Keep Moving!
 - a. EMT-Basics are provided 10 minutes to conduct a full patient assessment and begin treatment in the State EMT-Basic Practical Examination.
 - b. In an MCI, such lengthy patient assessments are not practical. S.T.A.R.T. assessments should last approximately 30 seconds per patient.

B. Starting the process:

1. Start the process as you enter the scene. Each patient should be triaged as the provider comes to them and should not be bypassed for a more severely injured patient.

C. Steps in assessment:

1. Step 1 is to move the green patients.
 - a. Make a clear verbal announcement that those who can walk should get up and do so to an easily recognized point away from immediate danger and outside the initial triage area.
 - b. These patients are prioritized as GREEN. These patients should be appropriately tagged and personnel should be assigned to supervise these patients
2. **RESPIRATIONS** assessment is Step 2
 - a. Open the airway, if necessary

- c. If there are **no** respirations - BLACK.
 - d. If respirations are **more than** 30 per minute - RED
 - e. If respirations are **less than** 30 per minute - go to step 3 (Perfusion)
3. Capillary refill or radial pulse check to assess **PEFUSION** is Step 3
- a. No radial pulse OR the capillary refill > 2 seconds - RED
 - i. Quickly attempt to control any severe bleeding and place the patient in shock position if necessary.
 - b. Radial pulse present OR capillary refill < 2 seconds – go to step 4
4. Step 4 is a check of **MENTAL STATUS**.
- a. If the patient has an **altered mental status** - RED.
 - b. If the patient has the **appropriate mental status** - YELLOW or GREEN according to other findings (obvious injuries or illnesses).
5. A copy of the START algorithm can be found in Attachment 1.
6. A copy of the JumpSTART algorithm, a triage algorithm for pediatric patients can be found in Attachment 2.

D. Continuing the Process

- 1. Triage is a continuous process that should be conducted multiple times on each patient.
- 2. Secondary triage is an in-depth reassessment based on clinical experience and judgment.
- 3. Continued triage can be done:
 - a. Prior to placement in the Treatment Area
 - b. While in the Treatment Area
 - c. On the way to the hospital

4. A Triage Tag can be placed on the patient at any point and can be changed to reflect a change in triage status.

VI. SALT TRIAGE

A. S.A.L.T. Development

1. In 2006 the Center for Disease Control (CDC) created a workgroup to conduct a review of triage methods. After a thorough literature review the workgroup created a triage system known as S.A.L.T. (Sort, Assess, Lifesaving Interventions, Treatment/Triage).
2. The CDC workgroup developed the Model Uniform Core Criteria (MUCC) for Mass Casualty Triage to help interoperability among existing triage tools. The MUCC consists of four general categories and 24 specific criteria which is recommended as the model minimum for all MCI triage systems.

B. General Considerations

1. Must comply with all ages and populations of patients
2. Must be applicable across a broad range of incidents
3. Must be simple, easy to remember, and amenable to quick memory aids
4. Must be easy to apply and practical for use in austere environments
5. Must be resource dependent and allow for dynamic triage decisions
6. Must have easily identifiable triage categories (tags, ribbon, etc).

C. Life-Saving Interventions

1. In the SALT triage method there are minimal life-saving interventions that should be done.
 - a. Bleeding Control
 - b. Opening the Airway
 - c. Chest Decompression
 - d. Auto-injector antidotes

2. Treatment should be completed, as needed, PRIOR to triage
3. Any necessary treatments should be complete only if
 - a. Equipment is readily available
 - b. Intervention is within the scope of practice of the provider
 - c. Triage can be performed within a minute

D. Categories

1. IMMEDIATE

- a. These patients are RED tagged
- b. An assessment of the patient will find the following:
 - i. Unable to follow commands or make purposeful movements or,
 - ii. Do not have a peripheral pulse or,
 - iii. Have obvious respiratory distress or,
 - iv. Have life-threatening external hemorrhage
 - v. Injuries are likely to be survivable given available resources

2. DELAYED

- a. These patients are YELLOW tagged
- b. An assessment of the patient will find the following:
 - i. Able to follow commands or make purposeful movement and
 - ii. Have peripheral pulses and
 - iii. Are not in respiratory distress and
 - iv. Do not have life threatening external hemorrhage and
 - v. Have injuries that are not considered minor

3. MINIMAL

- a. These patients are GREEN tagged

- b. An assessment of the patient will find the following
 - i. Able to follow commands or make purposeful movement and
 - ii. Have a peripheral pulse and
 - iii. Are not in respiratory distress and
 - iv. Do not have a life-threatening external hemorrhage and
 - v. Injuries are considered minor

4. EXPECTANT

- a. These patients are GREY tagged
- b. An assessment of the patient will find the following:
 - i. Unable to follow commands or make purposeful movements or
 - ii. Do not have a peripheral pulse or
 - iii. Are in obvious respiratory distress or
 - iv. Have a life-threatening hemorrhage and
 - v. Unlikely to survive given available resources

5. DECEASED

- a. These patients are BLACK tagged
- b. An assessment of the patient will find the following:
 - i. Not breathing after one attempt to open their airway (two rescue breaths given if the patient is a child)
- c. Normal resuscitative efforts are not appropriate in a mass casualty incident

E. Conducting SALT

1. Step 1: Global Sorting

- a. Must use simple commands for initial prioritization
- b. The first priority is those in need of life-saving interventions. Obvious life threats are those unable to follow commands and/or have no purposeful movements
- c. The second priority are those unable to ambulate but can follow commands and/or make purposeful movement.

- d. The third priority is able to follow commands and have no obvious life threats.
- e. All patients are assessed individually regardless of prioritization.

2. Step 2: Individual Assessment

- a. Individual assessment does not include vital signs and diagnostic equipment is not utilized.
- b. From the individual assessment patient prioritization falls within one of five categories. These are based on the injuries and assessment findings. Each category is designated by a different color.
- c. It is prior to the individual assessment that the minimal treatments occur. These include bleeding control, opening the airway, chest decompression, and auto-injector antidotes. The treatments should occur only if the equipment is available
- d. The first step in assessment is to determine if the patient is breathing. If they are not, attempt to open their airway. If the patient is still not breathing they are **DECEASED**.
- e. If the patient is breathing, determine if any of the following apply:
 - i. Can they covey commands or make purposeful movements?
 - ii. Do they have a peripheral pulse?
 - iii. Do they appear to be in respiratory distress?
 - iv. Do they have a major bleed?
- f. If they answer yes to any of the above and are not likely to survive given the resources they are **EXPECTANT**. If they are likely to survive given the available resources, they are **IMMEDIATE**.
- g. If the patient has an open airway and the answer to all of the below questions are “Yes” and their injuries are more than minor injuries, then, they are tagged **DELAYED**. If the patient has an open airway, the answers to the below questions are “Yes” and their injuries are considered minor, then, the patient is **MINIMAL**.
 - i. Can they covey commands or make purposeful movements?
 - ii. Do they have a peripheral pulse?
 - iii. Do they appear to be in respiratory distress?
 - iv. Do they have a major bleed?

3. Step 3: Treatment and/or transport
 - a. During this step, the ability to transport the patient and the order in which they are transported is determined.
 - b. First priority is given to **IMMEDIATE** patients.
 - c. Second priority is given to **DELAYED** and **MINIMAL** patients.
 - d. As resources allow **EXPECTANT** patients are transported after all others have been transported.
4. The full algorithm for SALT triage is located in Attachment 3.
5. Attachment 4 provides a slide-by-side comparison of START and SALT.

VII. THE VIRGINIA APPROVED TRIAGE TAG (ATTACHMENT 3)

A. Construction

1. Made with white weather resistant material - designed for use with a ballpoint pen.

B. Capabilities:

1. Multiple triage assessments of the patient.
2. Continuous patient information recording.
3. Continuous patient accountability and tracking.
4. Designed for easy interface with patient hospital records.

C. Format:

1. Front contains patient information section:
 - a. During MCIs the information not always obtainable.
 - b. Information can be added throughout triage, treatment, transportation and hospital reception phases
2. Triage Status section has space provided for four individual evaluations:
 - a. INITIAL - START assessment.

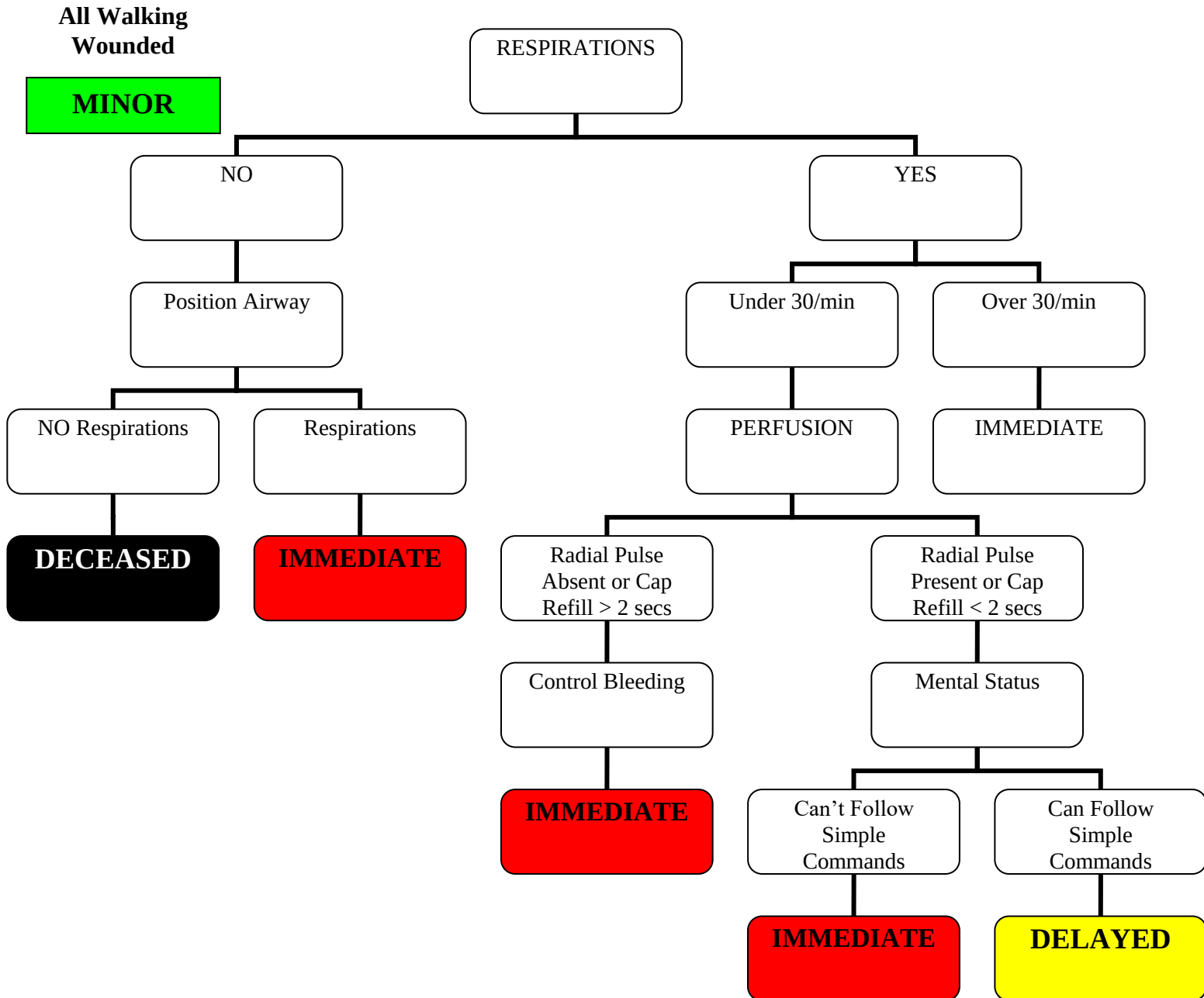
- b. SECONDARY - reassessment at the scene or in the treatment area.
 - c. BLANK - can be used in the treatment area or during transportation.
 - d. HOSPITAL - initial reassessment at the receiving hospital.
- 3. Chief Complaint section.
 - a. Major obvious injuries or illnesses can be circled.
 - b. Indicate injuries on the human figure.
 - c. Additional information is added on the Comments line.
- 4. Transportation Line. The transporting unit notes:
 - a. Agency information,
 - b. Destination hospital, and
 - c. The time the patient actually arrived.
- 5. Pull-off label section
 - a. A total of six labels are provided; two are marked for:
 - i. TREATMENT - to document on patient information worksheets.
 - ii. HOSPITAL - to tie the triage tag and scene patient number to the patient's hospital records.
 - iii. "Other" labels can be used for a variety of purposes, based on incident needs:
 - (a) Other tactical worksheet needs on scene, including transportation accountability.
 - (b) Marking personal effects.
 - (c) Use within the hospital.
- 6. Transportation Record section.
 - a. Detachable by tear off or as a pull off label.

- b. Used to document patients removed from the scene to a hospital or other facility.
 - c. Can be fixed to the transportation tactical worksheet - make certain hospital destination is marked.
- 7. Vital Signs section - space is provided to record three sets of vital signs.
- 8. Medical History section.
 - a. Information can be obtained any time during the incident.
 - b. Information can be obtained from Medic Alert identification devices.
 - c. Relevant medical history if known.
- 9. Treatment Record section.
 - a. Documents treatment sequence and progress.
 - b. Quick documentation of common treatments.
 - c. Space provided for additional treatments and remarks.
 - d. Spaces provided for time treatment actions are taken and for provider initials.

VI. SUMMARY

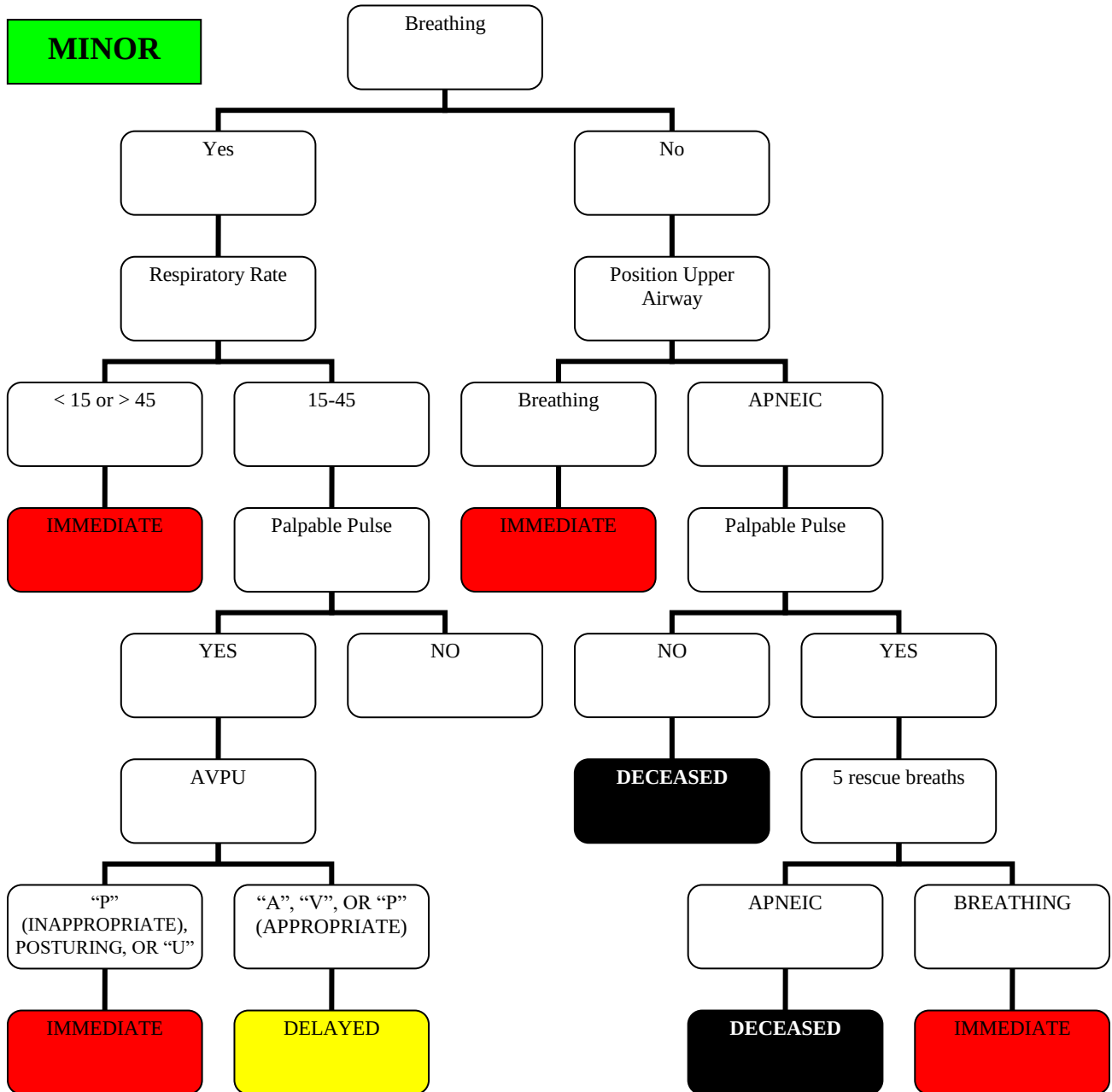
- A. Incidents of any kind have the potential to overwhelm EMS system personnel, equipment, resources, and medical facilities.
- B. It is necessary for providers to train and practice with agencies they respond with. Knowing what each agency utilizes and integrating with them is essentially in developing a coordinated response.
- C. Preparations and preplanning efficiency in all elements of mass casualty management

Attachment 1



**All Walking
Wounded**

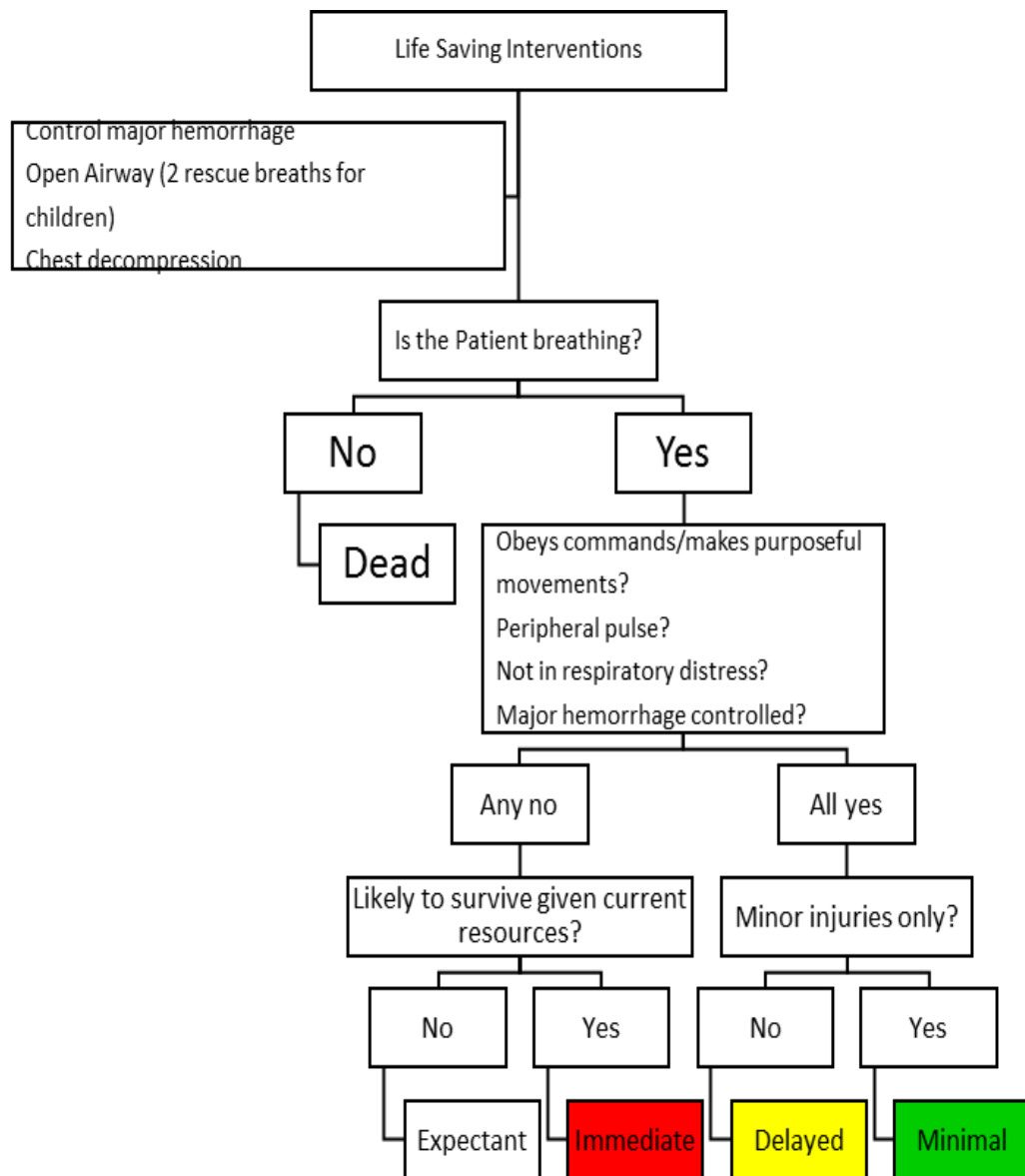
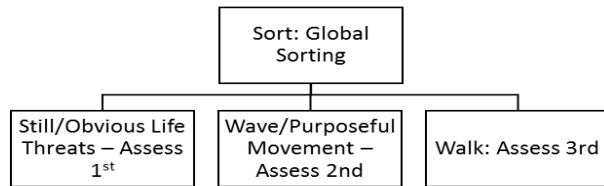
Attachment 2



*evaluate infants first in secondary triage using the entire JumpStart algorithm

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Attachment 3: SALT TRIAGE



Attachment 4. Comparison of START versus SALT Categories

IMMEDIATE	
START	SALT
Problems with: • R – Respirations • P – Perfusion • M – Mental Status • Severe burns that compromise the airway	• Unable to follow commands or make purposeful movements OR • Do not have a peripheral pulse OR • Obvious respiratory distress OR • Life-threatening external hemorrhage • Injuries are likely survivable given available resources

DELAYED	
START	SALT
• No significant problems to: ○ R – Respirations ○ P – Perfusion ○ M – Mental Status • Burn patients without airway problems • Able to follow commands, but unable to self-relocate to pre-designated GREEN area	• Able to follow commands or make purposeful movements AND • Have a peripheral pulse AND • No in respiratory distress AND • Do not have life-threatening external hemorrhage • Have injuries not considered minor

MINIMAL	
START	SALT
• “Walking wounded” • Able to self-relocate to predesignated GREEN area	• Able to follow commands or make purposeful movements AND • Have a peripheral pulse AND • No in respiratory distress AND • Do not have life-threatening external hemorrhage • Injuries are considered minor

DECEASED	
START	SALT
- Dead/non-salvageable (lowest priority) - Non-breathing patients - Resuscitation normally attempted - Not salvageable given resources available early in MCI response	- Not breathing after one attempt to open the airway - In children – two rescue breaths given - Normal resuscitative methods not appropriate in mass casualty

EXPECTANT	
START	SALT
- Not currently utilized in START triage - Compares to Yellow-Prime - No longer used in START triage	- Unable to follow commands or make purposeful movements OR - Do not have a peripheral pulse OR - Obvious respiratory distress OR - Life-threatening external hemorrhage - Unlikely to survive given available resources